



Solving archaeological puzzles

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ARTICLE INFO

Article history:

Received 31 August 2020

Revised 12 April 2021

Accepted 16 May 2021

Available online 27 May 2021

Keywords:

Re-assembly

Computer vision

Computer graphics

ABSTRACT

This paper focuses on the re-assembly of an archaeological artifact, given images of its fragments. This problem can be considered as a special challenging case of puzzle solving. The restricted case of re-assembly of a natural image from square pieces has been investigated extensively and was shown to be a difficult problem in its own right. Likewise, the case of matching “clean” 2D polygons/splines based solely on their geometric properties has been studied. But what if these ideal conditions do not hold? This is the problem addressed in the paper. Three unique characteristics of archaeological fragments make puzzle solving extremely difficult: (1) The fragments are of general shape; (2) They are abraded, especially at the boundaries (where the strongest cues for matching should exist); and (3) The domain of valid transformations between the pieces is continuous. The key contribution of this paper is a fully-automatic and general algorithm that addresses puzzle solving in this intriguing domain. We show that our approach manages to correctly reassemble dozens of broken artifacts and frescoes.

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1. Introduction

Puzzle solving has been an intriguing problem for many years. It has numerous application areas, such as in biology [1], shredded documents [2], image editing [3], and archaeology [4–6]. Even though the problem is NP-complete [7], different solutions have been proposed. The first computational solver was introduced in 1964 [8] and was able to handle a nine square-piece puzzle. More than five decades later, this problem is still as fascinating and intriguing. State-of-the-art techniques mostly address natural images and are based on color matching [9–17], on shape matching [18–24], or on a combination of both [25–31].

We focus in this paper in the domain of cultural heritage, which has been an important application of computer vision & computer graphics for many years [32–34]. The vast majority of archaeological objects are discovered in a fragmentary and poor state, which hampers the extent of archaeological research that is possible to do on them. Much effort and time is expended in trying to manually reassemble these fragments as accurately and completely as possible.

We concentrate on archaeology not only because cultural heritage has been acknowledged worldwide as an important goal, but also because the archaeological domain exposes the limits

of current computational techniques. Archaeological artifacts are not “clean” and “nicely-behaved”; rather they are broken, eroded, noisy, and ultimately extremely challenging to algorithms that analyze or reassemble them. Therefore, from the computational point of view, archaeology serves as an extremely challenging application area.

A few re-assembly algorithms have been proposed for this domain. They focus on 3D puzzle problems of non-negligible thickness [6,35–37]. Conversely, we consider 2D colored archaeological objects of planar or nearly-planar “thin-walled” pieces. Our problem is important for two reasons: First, there are many cases where this is the actual input, especially for frescoes and wall paintings. Second, the data enjoys scanning efficiency, simplicity, and easier utilization of color.

Three major properties characterize fragments of colored archaeological artifacts. These properties result from abrasion, color fading, and continuity (Fig. 1). Abrasion creates gaps between the pieces, which makes it harder to match adjacent fragments. Color fading results in spurious color edges. These should be distinguished from real edges and gradients, which are highly important for matching. Lastly, the valid transformations between every pair of fragments belong to a continuous space. This raises many questions: what is a valid transformation? how shall we sample the space of rigid transformations? how can we compute matching scores, knowing that the transformation is imprecise due to sampling?

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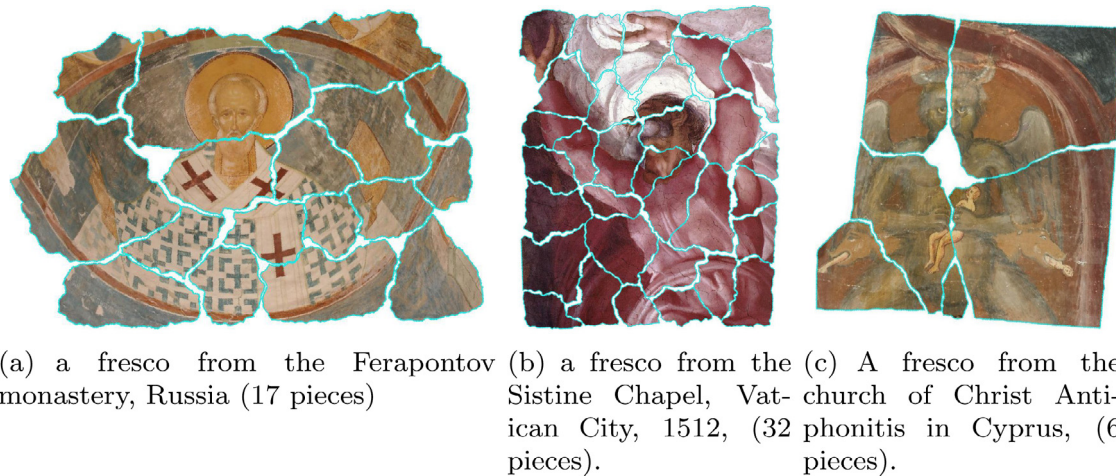


Fig. 1. Fragments are re-assembled by our algorithm. The borders of the fragments are marked in cyan. The fragments are faded, stained, scratched, and the boundaries do not accurately match.

We propose a novel re-assembly algorithm that handles these challenges. It is based on four key ideas. First, in order to address fragment abrasion, we propose to extrapolate each fragment prior to re-assembly. This reduces the continuity problem (predicting how to “continue” the fragment) we are facing into a matching problem. Second, we suggest a transformation sampling method, which is based on the notion of *configuration space*, and is especially tailored to our problem. Third, at the core of any puzzle solving algorithm lies the question of what makes a good match. We propose a new measure, which takes into account the special characteristics of the domain: the gaps between the pieces, color fading, spurious edges, the varying length of the matching boundary, and the imprecise transformations. Finally, our placement considers not only the above scores, but also our confidence in the match. This is important because even a good matching score does not necessarily indicate adjacency. Therefore, our confidence is influenced by the uniqueness of the match and the fragment size.

Our algorithm has been tested on real objects and frescoes from the British Museum collections, as well as on images of frescoes from various churches world-wide whose fragmentation was artificially generated in a realistic manner. We demonstrate state-of-the-art results on these examples.

Hence, this paper makes several contributions:

1. First and foremost, it proposes a novel approach for solving puzzles whose pieces are of general shape, which need not necessarily match precisely due to erosion, and which have degraded unevenly over time.
2. We define the notion of *valid transformations* and propose a method of sampling them. This method may find uses in other domains.
3. We define a novel dissimilarity score that takes into account the special characteristics of the domain.
4. We introduce the notion of confidence in the dissimilarity scores, which depends not only on the value of the dissimilarity, but also on those of the competing matches. Confidence may be used not only in re-assembly, but also in other applications that are based on dissimilarity.

2. Related work

Fragmented object reconstruction has been considered challenging in a variety of areas, including puzzle solving [38], recon-

structing torn documents [2], image editing [3], and reassembling fragmented artifacts [4–6]; see [35,39] for excellent reviews.

Re-assembly in Archaeology. Finding a way to bring ruined ancient artifacts back to life has sparked the imagination of scientists for many years [35]. Most of the work has been done on 3D artifacts. Two facets of the problem have been studied: finding good features that would indicate a match between a pair of fragments and proposing global reassembly methods.

Generating a set of pairwise matches is based on a variety of properties. These include color (when exists) [37], 2D fragment contour shape [19,40,41], or fracture surface alignment [4,42–44]. Supervised learning approaches that aggregate many properties have also been used to predict fragment matches [36,37]. In particular, given two scanned 3D fragments and a proposed aligning transformation, the probability that the match is correct is generated in [36], by learning this probability. Deep learning has been used in [45] for matching archaeological fragments which do not have to be square. The algorithm however does not deal with gaps between the fragments.

Once matches have been computed, global reassembly algorithms aim to select a subset of matches that would place the fragments consistently. Various heuristic approaches have been proposed. For instance, hierarchical clustering has been used in [19,42,46,47]. Greedy approaches have shown to be beneficial in [21,41,48]. In [6] a genetic algorithm is introduced, where a highly challenging archaeological example is solved; the features introduced in [36] were utilized.

The current paper handles both sub-problems, the local and the global, albeit in 2D. As features, it uses the colors directly and the geometry indirectly, in order to be able to handle the gaps between the pieces. We pursue a greedy re-assembly algorithm, which is adapted to our problem.

Handling 2D general-shaped pieces and gaps. Though not in archaeology, the most closely related works to ours are [31,49], which assume general-shaped pieces of natural images. However, in [31] there are no gaps between the pieces. In [49], the focus is on the gaps, but the user must provide an approximate initial placement. Our algorithm successfully solves [49]’s examples, but without requiring an initial placement. Furthermore, it can handle the more challenging images of archaeological artifacts.

The most recent work in a domain that is related to ours is [50]. This work concentrates on the gaps between the pieces and manages to deal with large gaps. However, it is assumed that all the pieces are square and that all the puzzles are only 3×3 . This

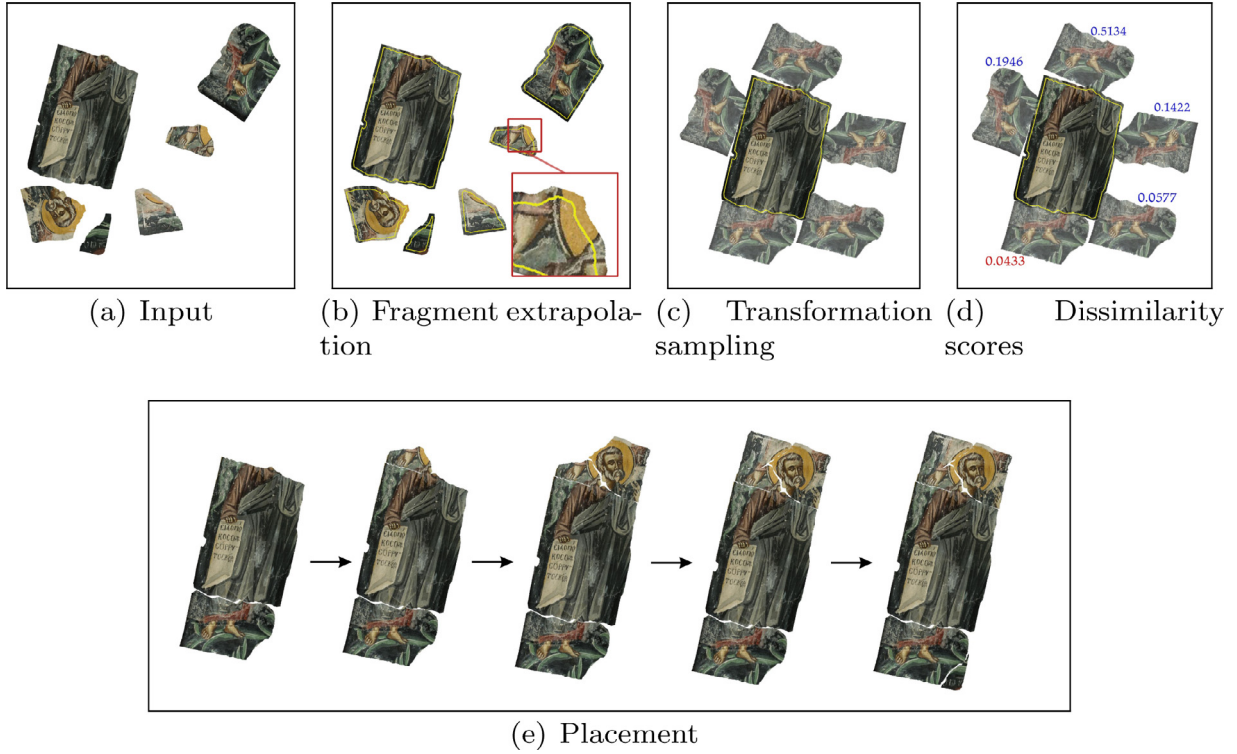


Fig. 2. Algorithm outline.

method cannot be easily generalized to handle more pieces due to its exponential complexity.

To summarize, the problem we handle differs from the above works in various aspects—shape, size, type, or dimension.

3. Algorithm outline

Given N images, each capturing a single fragment of an artifact, our goal is to reassemble the artifact. In particular, each fragment should be assigned a 2D rigid transformation that indicates how to place it within the re-assembly. Our proposed algorithm consists of four steps, as illustrated in Fig. 2. We briefly discuss each of the steps below and elaborate on Steps B–D in subsequent sections.

A. Fragment extrapolation. In the first step, each fragment is extrapolated. This serves two goals. First, it shows how the fragment used to look before it was damaged and eroded. Second, it reduces the problem of continuity from one fragment to another into the problem of matching pieces within the extrapolated region.

Formally, given a fragment, the goal is to produce a k -pixel wide band around it. Several approaches have been proposed to extrapolation [51–55]. We adopt the example-based approach of [56], which manages to produce sharper and relatively more accurate extrapolations for our examples. The algorithm extrapolates the image using patches from the original image, utilizing patch-based optimization. This optimization takes into account not only colors and image gradients, but also geometric and photometric transformations, which is vital for our case. The optimized energy is based on mixture of an L_2 norm for colors and an L_0 norm for gradients, which are sparse. Moreover, in the search for similar patches, the method compensates for both geometric (translation, rotation, non-uniform scale) and photometric transformations to address structure align-

ment as well as for color inconsistencies. This results in a gradual transition with less sacrificing texture sharpness.

One of the consequences of this method is that only the colors appearing in the original image will appear in the extrapolated image. Thus, the number of colors in the image stays the same. We exploit the fact that in frescoes the number of colors is small, as will be explained in Section 5.1.

Fig. 3 shows some of our extrapolation results, where the width of the extrapolated strip depends on the fragment size (1.6% of the bounding box perimeter). The results manage to continue the colors and the structure of the image in most cases. It is interesting to note that despite of the imperfect extrapolation of Figure 3(d), our algorithm manages to solve the relevant puzzle flawlessly (Figure 2).

B. Sampling a set of valid transformations. The goal of this stage is to sample the space of transformations between a given pair of fragments. This sampling should satisfy three conditions: First, the transformations must be valid. For that, we need to define the concept of validity in our context. Briefly, we wish the transformed fragments to be close to each other (so that the likelihood of adjacency is high), but to not to overlap. Therefore, we require that the pixels of the original fragments do not overlap, but that the extrapolated pixels of the fragments do. Second, we want the number of sampled transformation to be feasible. To achieve this, we show how the set of valid transformation can be described as a configuration space problem, used in robotics [57]. This space can be more easily sampled. Finally, due to sampling, the promising transformations might be inaccurate. A way to fine-tune the transformations is needed. We will show how this can be performed during placement. We elaborate on this step in Section 4.

C. Finding good matches. Computing a dissimilarity score for a given transformation between a pair of fragments lies at the core of re-assembly. Thanks to Step A, the continuity problem was reduced to matching the extrapolated region of one frag-

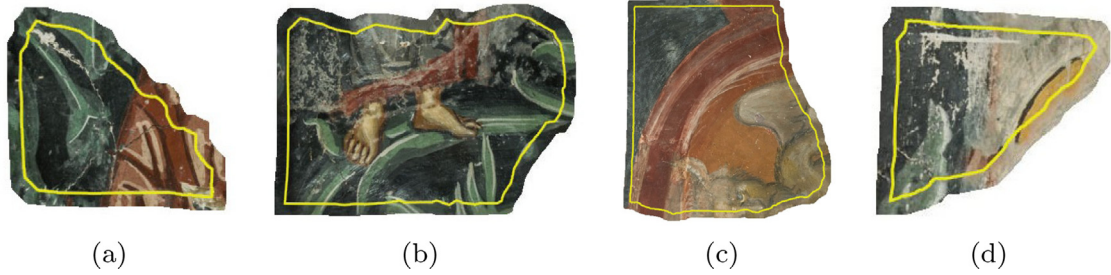


Fig. 3. Extrapolation results. The original boundaries are marked in yellow. Note how the patterns continue nicely in (a)–(c). However, in (d) the yellow circle does not continue as expected, due to lack of similar patches. Nevertheless, our re-assembly algorithm solves the puzzle flawlessly. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

ment and the other original fragment, on the overlapping region. The strongest cues for matching are the colors and the color edges, which should continue from one fragment to its adjacent piece. Matching becomes challenging in our domain due to three additional difficulties: (1) faded colors (2) imprecise transformations (due to sampling) cause misalignment of the edges and (3) degradation may yield spurious edges that should be ignored. Our algorithm handles these problems. Even when the dissimilarities are computed, our problem is not yet solved. This is so since a low dissimilarity value does not necessarily imply confidence in a match, as other matches with the same fragment may be almost as likely. Therefore, we propose to compute a confidence score, which predicts the likelihood of the pair of fragments to be neighbors. The confidence in the match takes into account, in addition to the dissimilarity score, also the scores of competing matches in the same region of the fragment. This step is explained in detail in Section 5.

D. Placement. Similarly to most previous work, our placement strategy is greedy, iteratively adding one (“best”) fragment a time. New complications that are due to both the continuity of the space of transformations and the general shape of the fragments, need to be efficiently addressed. Furthermore, to handle the problem of the imprecision of the transformation, the selected transformation should be refined in a small neighborhood around it. We address these issues in Section 6. For a demonstration of the placement algorithm, we refer the reader to its visualization on YouTube [58].

4. Sampling valid transformations

Given a pair of fragments, the goal is to efficiently find the best transformation between them. To reach this goal, we proceed in three stages. First, we define the notion of *valid* transformations. Then, we propose an efficient algorithm for validating a given transformation. Finally, since computing the dissimilarity scores for all valid transformations is intractable, as the space of valid transformations is continuous, we introduce a method for sampling this space.

4.1. Valid transformations

We are given two fragments f_i, f_j , and their extrapolations f_i^+, f_j^+ . A legal transformation is one where f_i and f_j do not overlap. In addition, the archaeologists require that the two fragments should be quite close to each other. Therefore, we define a transformation T to be valid if the extrapolated fragment f_i^+ overlaps the transformed fragment $T(f_j)$, but the original fragment f_i does not overlap it. Formally, T should satisfy the following constraint, as illustrated in Fig. 4:

$$T(f_j) \cap f_i^+ \neq \emptyset \wedge T(f_j) \cap f_i = \emptyset. \quad (1)$$

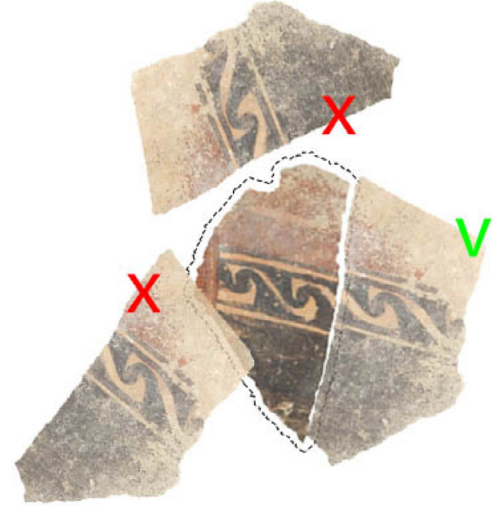


Fig. 4. Valid transformations. The fragment must overlap the extrapolated strip, but not the original fragment (the extrapolation is marked with a dotted line). Marked in green is a valid transformation, and in red are two invalid transformations—one due to intersection with the fragment and the other due to not intersecting the extrapolation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.2. Efficient transformation validation

Verifying the validity (Eq. (1)) of every sampled transformation, by testing for polygon overlap, is very expensive. The key idea of bypassing this procedure is to borrow the concept of a *configuration space* from the field of robotics [57]. This reduces the problem of checking for overlap between fragments to the problem of checking whether a point is inside a polygon.

Briefly, given a translating and rotating robot $\mathcal{R}$ and an obstacle $\mathcal{P}$, the configuration space is a 3-dimensional space $\mathbb{R}^2 \times [0 : 2\pi)$, which is defined as follows. A point $p = (x, y, \theta)$ in the configuration space corresponds to a certain placement of the robot in the 2D plane, after rotating the robot by θ and translating it by (x, y) . The configuration space is defined as the set of points such that the corresponding placement of $\mathcal{R}$ after rotation by θ intersects $\mathcal{P}$. This can be efficiently computed by the Minkowski sum operator $\oplus$ as:

$$\mathcal{C}(\mathcal{P}, \mathcal{R}, \theta) = \mathcal{P} \oplus -\mathcal{R}(0, 0, \theta), \quad (2)$$

where $\mathcal{R}(0, 0, \theta)$ is the robot's location after rotation by θ and no translation. Once the Minkowski sum has been computed as a polygon, we can easily test whether a point is in the configuration space (i.e., inside the polygon). The configuration space is computed for each θ independently.

The free space (free of intersection) is the complementary of Eq. (2), i.e.,

$$\mathcal{F}(\mathcal{P}, \mathcal{R}, \theta) = \overline{\mathcal{C}(\mathcal{P}, \mathcal{R}, \theta)}. \quad (3)$$

In our case, we wish to convert Eq. (1) to Eqs. (2)-(3). Let fragment f_j be the robot and f_i & f_i^+ be the obstacles. We require that $T(f_j)$ will not intersect f_i (free space), but will intersect f_i^+ (configuration space). Therefore, we can formulate our validation operator $\mathcal{V}$ as

$$\mathcal{V}(f_i, f_j, \theta) = \mathcal{F}(f_i, f_j, \theta) \wedge \mathcal{C}(f_i^+, f_j, \theta). \quad (4)$$

Eq. (4) must be computed also when the roles of f_i and f_j are switched, for the inverse transformation. A given transformation is valid when both conditions hold.

4.3. Sampling

Once we have an efficient method for validity testing, sampling can be performed in three simple steps, as follows. First, N_r rotations (angles) are sampled uniformly. Then, for each sampled rotation, we construct two translational configuration spaces, one for f_i, f_j and the other for f_i^+, f_j , resulting in two polygons. The two polygons are sampled uniformly at resolution R_t , and transformations that satisfy Eq. (4) are maintained.

In our experiments we set $N_r = 80$ and R_t to 5 pixels in each direction. This suffices to find a high-score transformation in the neighborhood of the correct transformation. Later, during placement (Section 6), local refinements of the recovered transformations are performed by dense sampling around the proposed transformation, yielding accurate results.

4.4. Complexity

Configuration space construction is linear in the size of the fragments (assuming convexity); this is performed for each orientation, i.e., N_r times. Then, for each sampled translation a point-in-polygon procedure is performed; this is linear in the size of the polygon.

The straightforward alternative to our method is to check the overlap between every pair of fragments for each transformation, which is highly expensive. We note that another possible alternative approach is to find a globally optimal transformation using a branch-and-bound strategy [59]. In our case, this procedure is impractical, both due to its complexity and due to the difficulty to incorporate Eq. (4), as well as our dissimilarity function, into that framework.

5. Finding good matches

A good *dissimilarity* score is a fundamental component in every jigsaw puzzle solver. It predicts the likelihood of two aligned fragments to be neighbors. However, a dissimilarity score by itself provides only partial information. Even if two fragments are very similar, other fragments may also have good scores and compete for the same region of the fragment. Which of the similar matches should be chosen? To consider competition, we add another measure, which indicates our *confidence* in a match. Below, we describe these two scores, dissimilarity and confidence.

5.1. Dissimilarity

Given two pieces, f_i, f_j and a transformation T , we are seeking a dissimilarity measure $d(f_i, f_j, T)$ that predicts the likelihood of f_i and f_j to be neighbors, after transforming f_j by T . What would make a good match? First, the patterns and the colors of f_i should continue seamlessly to $T(f_j)$. This is especially important

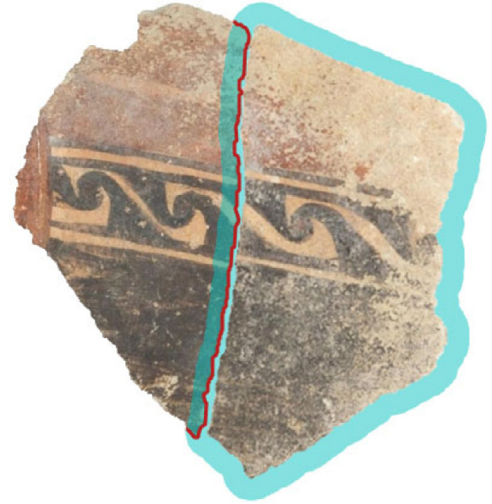


Fig. 5. Overlap curve. $\Psi_{i,j}^T = \partial f_i \cap T(f_j^+)$, in red, is the curve defined as the overlap of the boundary of f_i (left) with the extrapolation of the transformed f_j (right). Note the slight misalignment of the edges. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

near the gradients, as edge continuation is a strong hint for correct re-assembly. Second, the length of the match should be considered; the longer the match, the higher the probability that the match is correct and not random.

Our algorithm is based on three key ideas. First, we view the extrapolation as a prediction of the continuation of the fragment. Therefore, it suffices to compare the overlap of the vicinity of the boundary of one fragment with the extrapolation of the other. Second, we look for the continuation of image gradients between the pieces. However, since due to erosion, there are superfluous gradients, we discard the latter using knowledge about our domain. Third, even a slight error in the transformation between the fragments causes the color gradients to be misaligned. Since our transformations are sampled, we ignore slight misalignment. We elaborate on the realization of these ideas hereafter.

Let $\Psi_{i,j}^T = \partial f_i \cap T(f_j^+)$ be the curve defined as the overlap of the boundary of f_i with the extrapolation of the transformed f_j ; see Fig. 5. This curve defines a set of pixels both in f_i and in f_j^+ , which should be compared.

To accommodate the first idea, we compute the color difference for the pixels on $\Psi_{i,j}^T$, as follows:

$$d_c(f_i, f_j^+, T) = \sum_{p \in \Psi_{i,j}^T} D(f_i(p), T(f_j^+)(p)), \quad (5)$$

where $D(u, v)$ is the L_1 distance between the colors of pixels u and v , in the LAB color space.

Inline with the second idea, we would like to disregard the gradients that are due to erosion. We make use of the fact that archaeological artifacts use only a few colors (which are culture-dependent). We therefore generate a *color palette* to represent the original colors [60]. A palette is a small set of colors that represents the full range of colors in the image. The palette is constructed using a variant of the k -means algorithm, where k is the number of colors. As k is unknown, we compute palettes for different values of k ($k = 3, 4, 6, 8, 10$). The principle is that the smaller the value of k in which the gradient appears, the more important this gradient is.

Before explaining how the different palettes are utilized, recall that due to the inaccuracy of the transformations, the gradients are slightly misaligned. To jointly realize the second and the third ideas, we build a weighted histogram of the gradient directions. In

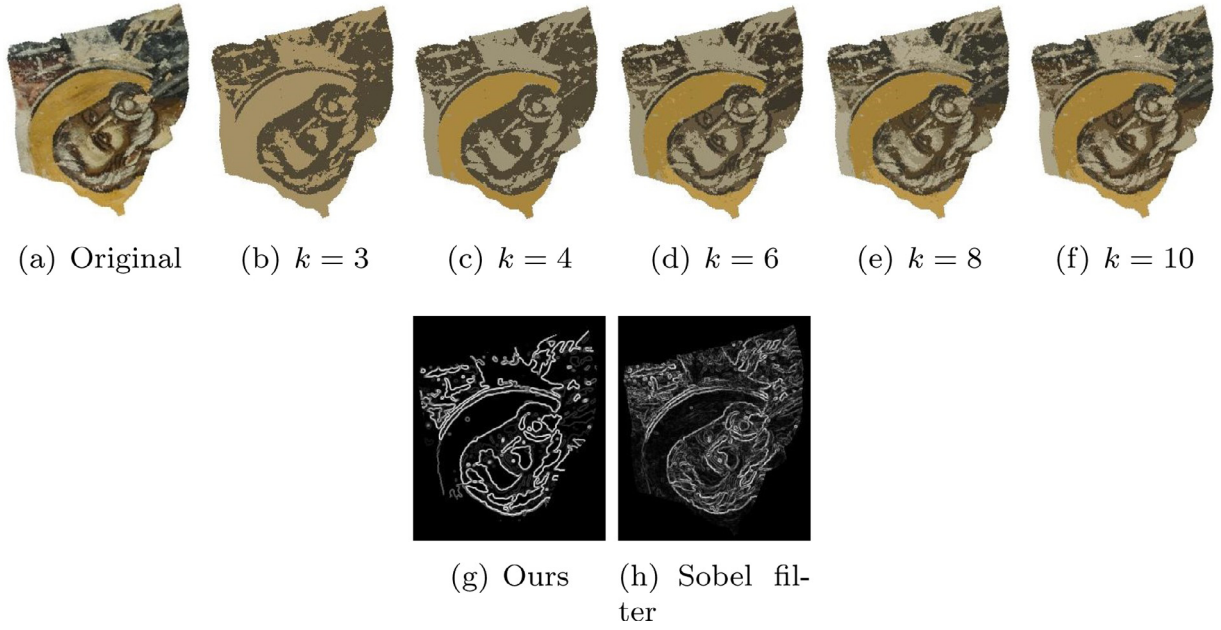


Fig. 6. Palettes for gradient computation. As the number of colors (k) in the palette grows, more details become visible. Gradients that appear early on and remain consistent across many palettes, are more important and hence are weighted stronger. The result (g) is sharper and lacks superfluous edges compared to (h).

this histogram, slight misalignments do not affect the score. Furthermore, each gradient is weighted by $1/k$. Thus, the weight of a gradient is high when it is repeated in many palettes and when it appears early on, in palettes consisting of a small number of colors.

Fig. 6 demonstrates this method. As we add more colors to the palette, the image is further segmented, showing more details. Yet, the edges that are due to erosion do not appear. Furthermore, the important edges appear in all palettes, hence they are stronger. Contrary to classical edge detection, these are the only apparent edges.

We are given two histograms of gradients on $\Psi_{i,j}^T$, h_i, h_{j+} , for f_i, f_{j+} , respectively. These histograms are smoothed with a Gaussian kernel in order to deal with matching gradient directions that might fall in adjacent bins. The difference between the histograms is computed as follows (where d is a histogram bin index):

$$d_h(f_i, f_{j+}, T) = \sum_d |h_i(d) - h_{j+}(d)| (1 + \ln(a(d))). \quad (6)$$

Inspired by [61], we give more weight to larger bin differences; the latter is weighted by $1 + \ln(a(d))$, where

$$a(d) = \frac{\max(h_i(d), h_{j+}(d))}{\min(h_i(d), h_{j+}(d))}. \quad (7)$$

Putting it all together. The combined dissimilarity score takes into account both Eqs. (5) and (6), as well as the length of the intersecting curve $\Psi_{i,j}^T$. It is defined as:

$$d(f_i, f_{j+}, T) = \frac{d_h(f_i, f_{j+}, T) + d_c(f_i, f_{j+}, T)}{|\Psi_{i,j}^T|^\alpha}, \quad (8)$$

where the parameter α controls the importance of this length ($\alpha = 2$ is used in all our examples). The dissimilarity score, which indicates the quality of the match, is low when the color difference and the histogram difference are small, and the intersection curve is long.

Note that one can view the LAB values and the histogram of gradients on the boundary as a descriptor of the fragment. When a suggested transformation is given, such a descriptor is also computed in the extrapolated region of the other piece. From these two descriptors the dissimilarity score is computed.

An interesting property of our algorithm is that geometry is not addressed explicitly. That is to say, we do not attempt to match the boundaries by their shape. This is because we assume erosion and therefore shape matching is unlikely. Instead, our method performs “fuzzy”-matching, considering the geometry only implicitly. This is done by setting the location where pieces can match by the extrapolation and enforcing this location by the definition of the valid transformations. In addition, the dissimilarity function favors long intersecting curves $\Psi_{i,j}$ (Eq. 8), which would be the case when the geometry of the boundaries are similar.

5.2. Confidence

Given a pair of fragments f_i, f_j and a dissimilarity score of a given transformation T between them, our goal is to evaluate our confidence in this dissimilarity score. When would a small dissimilarity score indeed indicate adjacency? If there exist other candidate transformations that have very good scores on the same region within f_i , we cannot be confident in any of these good scores. Conversely, our confidence should be high when a match is unique, i.e. when only one transformation corresponds well in that region.

The key idea is to compare T 's dissimilarity with those of the competing transformations and to use this comparison to determine T 's uniqueness. To do that, we shall first define what the competing transformations are. We note that they can be associated either with f_i & f_j or between f_i and some other fragment. In the first case, we have to disregard T 's nearby transformations, as they would naturally give similar dissimilarity scores.

5.2.1. Disregarding nearby transformations

To disregard nearby transformations, we have to define the notion of distance between transformations. One could consider to measure the distance in transformation space, but since we are interested in matches, we propose a distance that suits our problem.

Given two valid transformations T_1, T_2 applied to f_j , we consider $\Psi_{i,j}^{T_1}, \Psi_{i,j}^{T_2}$, the two curves of intersection within f_i , after applying the transformations to f_j . (Note that the curve might be fragmented.) We compute the *Intersection over Union (IoU)* between the curves and define it as the closeness measure between

transformations T_1 and T_2 :

$$XfSim_{i,j}(T_1, T_2) = \frac{\Psi_{i,j}^{T_1} \cap \Psi_{i,j}^{T_2}}{\Psi_{i,j}^{T_1} \cup \Psi_{i,j}^{T_2}}. \quad (9)$$

The larger the overlap, the higher this value. Hence, if this value is higher than a predefined threshold γ_H , the transformations are considered close to each other ($\gamma_H = 0.85$ in our experiments). In this case, they are not considered competitors. The same condition is also tested when the roles of f_i and f_j are reversed. Hence, T_1 and T_2 are regarded as competitors if

$$(XfSim_{i,j}(T_1, T_2) < \gamma_H) \vee (XfSim_{j,i}(T_1^{-1}, T_2^{-1}) < \gamma_H). \quad (10)$$

5.2.2. Confidence computation

The key idea is to look for the most competitive match on $\Psi_{i,j}$ within f_i . If the dissimilarity of this competitive transformation is much worse than the score of T , then our confidence in T is high.

Specifically, we constructively define the set of *competitive transformations* as follows: We apply Eq. (9) to the curves of every sampled transformation of the pair (f_i, f_m) , $\forall m$. Every transformation S for which $XfSim_{i,m}(T, S) > \gamma_L$ is considered competitive. Intuitively, this condition implies that the overlap between the matches, defined by transformations S and T (of some fragment with fragment f_i), is high. We note that for $m = j$, T and S should also not be nearby (as defined in Eq. (10)).

Now that a set of competitive transformation is defined, we proceed to define our confidence $C(f_i, f_j, T)$ in transformation T . We choose from this set the best match with f_i , $competitor_i(T)$ i.e., the one with the lowest dissimilarity score. If the best competitor and T have similar scores, we are not confident that T is indeed the correct match. However, if their dissimilarities to f_i differ greatly, our confidence in the match is high. This is expressed in the following definition of the confidence:

$$\hat{C}(f_i, f_j, T) = 1 - \frac{d(f_i, f_j^+, T)}{d(f_i, f_m^+, competitor_i(T))}. \quad (11)$$

To further increase our confidence in a match, we recall the concept of *best buddies* [14]. Intuitively, two pieces are best buddies if they both agree that the other piece is their most likely neighbor. Thus, we would like to consider also the inverse transformation. However, since our transformations are sampled, the inverse of T might not have been sampled. To solve this, we define our confidence in matching two fragments f_i and f_j to be high if there exist two transformations, T_1 and T_2 , such that (1) T_1 transforms f_j about f_i with large confidence and (2) T_2 , which is “almost” inverse to T_1 , transforms f_i about f_j and also has high confidence. This is expressed as follows:

$$C(f_i, f_j, T_1, T_2) = \left(\frac{\hat{C}(f_i, f_j, T_1) + \hat{C}(f_j, f_i, T_2)}{2} \right), \quad (12)$$

where “almost” inverse is defined as

$$(XfSim(T_1, T_2^{-1}) \geq \gamma_H) \wedge (XfSim(T_1^{-1}, T_2) \geq \gamma_H).$$

6. Placement

The last phase of the algorithm is the re-assembly of the artifact from its fragments, given the scores previously computed. The complexity of the problem is extremely high, as given N fragments, there are $\binom{N}{2}$ pairs, and for each of them an infinite number of transformations (from which we sampled a finite set). Our goal is to find N transformations that will result in the correct re-assembly.

We follow the most common strategy in puzzle solving—greedy re-assembly [10–14], which iteratively selects the fragments to be placed. At every iteration, rather than looking for a fragment that

best matches a single previously-placed fragment, we look for one that maximizes the confidence with respect to multiple-placed fragments.

Prior to placement, we construct for each pair of fragments a set of the best (dissimilarity-wise) non-competing transformations between them. For each transformation in this set, its confidence value is computed, as explained in Section 5.

The placement is initialized by choosing the pair of fragments with the maximal confidence $C(f_i, f_j, T_1, T_2)$, for some transformation pair T_1, T_2 (Eq. (12)). Then, we iterate on the following three steps, on which we elaborate below:

1. **Discarding illegal placements:** Illegal (physically-impossible) transformations between placed and unplaced fragments are discarded.
2. **Updating the confidence:** The confidence values of candidate transformations, which match multiple-placed fragments, are re-calculated.
3. **Placement:** The next best piece is placed, after refining its corresponding transformation.

Let P be the set of the already-placed fragments at step k , where each fragment $f_i \in P$ is associated with transformation T_i . In Step 1 we discard illegal transformations, which would place a fragment f_j on top of a fragment from P . To efficiently verify whether $T(f_j)$ is legal, we use the configuration space that was already calculated during transformation sampling (Section 4). For T to be legal, we require that it does not belong to the configuration space of $\mathcal{C}(T_i(f_i), f_j)$ for all $f_i \in P$.

How can this be determined efficiently? We have only computed $\mathcal{C}(f_i, f_j)$ before, whereas we seek to know $\mathcal{C}(T_i(f_i), f_j)$. The trick we use is noting that the condition that $T \notin \mathcal{C}(T_i(f_i), f_j)$ is equivalent to the condition that $T_i^{-1}T \notin \mathcal{C}(f_i, f_j)$ and that $\mathcal{C}(f_i, f_j)$ was already computed.

Once a partial placement exists, the confidence values of a given transformation might no longer be valid and therefore must be re-calculated. This is the goal of Step 2. This requires us to first detect the neighbors of a transformed fragment and then calculate the dissimilarity & confidence values with the already-placed pieces. A given transformation is *valid* with regard to multiple-placed fragments, if it is valid (Eq. 1) for at least one placed fragment and legal for the rest of them. This means that we can use the previously-computed configuration spaces, as done in Step 1.

The dissimilarity and the confidence can now be re-computed. The key idea is to consider all the already-placed fragments as a single (virtual) fragment, f_p , and compute these values as described in Section 5.

Finally, in Step 3, the fragment f_j with the highest confidence is placed using the corresponding transformation T_j . The latter is now refined, by densely re-sampling the space of transformations in a small neighborhood around T_j , in order to address the inaccuracy of the transformation due to sampling. Then, the dissimilarities of the new sampled transformations to f_p are calculated.

The final transformation $\tilde{T}_j$ is the one that minimizes:

$$\tilde{T}_j = \arg \min_{T \in \mathcal{N}(T_j)} [d(f_p, f_j^+, T) + d(f_j, f_p^+, T)], \quad (13)$$

where $\mathcal{N}(T_j)$ is the set of transformations in the neighborhood of T_j . Due to the dense sampling an accurate transformation is found.

We note that the final assembly may change, depending on the initial transformation sampling. This might result in missing correct matches. To solve this, we could either increase the sampling resolution or perform the algorithm multiple times on randomly-rotated fragments and choose the best assembly. As the second solution is cheaper, we do it and run the algorithm five times. Then, our algorithm automatically chooses the best re-assembly, which is measured by its overall (sum of) confidence.

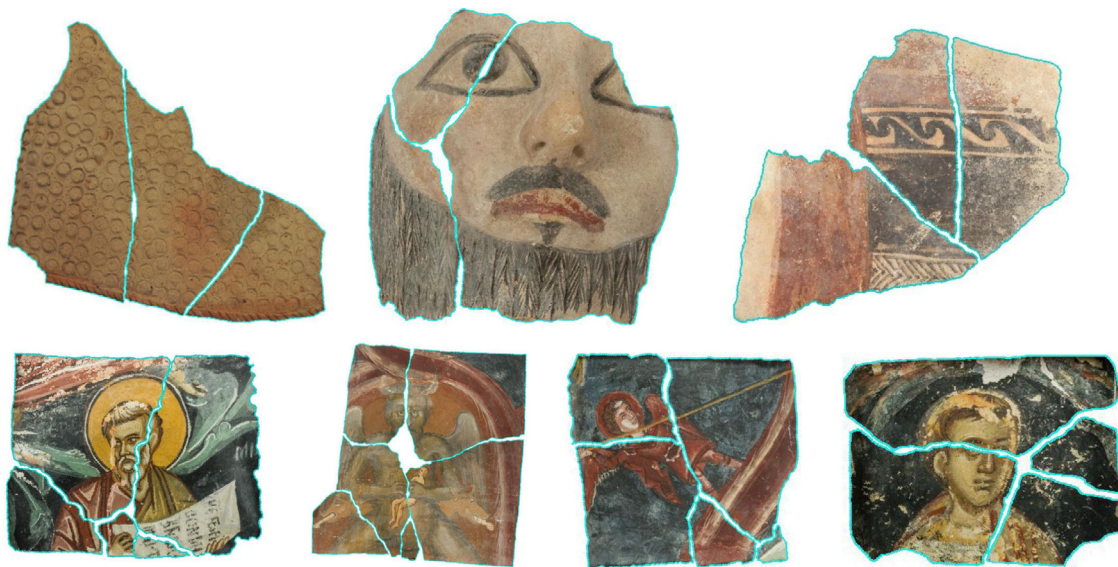


Fig. 7. Re-assembly of real artifacts. The top examples are 3D statues from Salamis, while the bottom examples are frescoes from the church of Christ Antiphonitis in Cyprus. The fragments are faded, stained, scratched, and with other apparent degradation artifacts. Furthermore, they do not accurately match due to erosion or missing pieces. Our algorithm manages to re-assemble them perfectly.

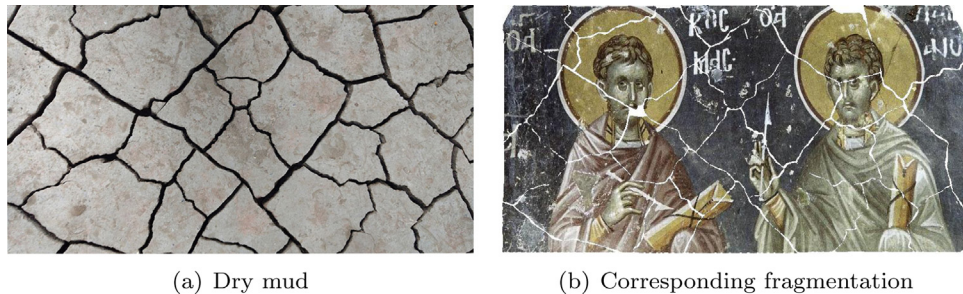


Fig. 8. Fragment generation. To generate “natural” fragments, we break a fresco based on a fragmentation of dry mud.

7. Results

This section describes the results of our algorithm, applied to archaeological artifacts. In some cases, the fragments are naturally broken, while in others, they were generated to emulate the way artifacts have been broken.

7.1. Qualitative results

Fig. 7 shows the re-assembly of some real artifacts. Some of the examples include images of fragments of 3D statues, whereas others are images of frescoes. These examples have eroded fragment boundaries, missing pieces, and faded colors. Some have very few colors, while others have more; moreover, the same colors appear in different regions of the artifact. It can be seen that our algorithm re-assembled all the artifacts flawlessly.

Since we could not obtain artifacts of this type with more fragments, the challenge was to generate additional challenging examples which are plausible, natural, and provide ground truth that will enable evaluation. In 3D, [62] proposes a heuristic that relies on *cutter objects* rather than on Finite-Element simulation. Simulating weathering effects remains an intriguing problem.

We propose a different and novel approach to generate plausible fragmentation, where erosion is naturally included. The key assumption is to borrow fragmentation examples from nature itself. Thus, given an image of a real artifact/fresco, we break it according to a pattern of dry mud, as illustrated in Fig. 8. Dry mud not only

breaks in a natural manner, but also generates gaps between the pieces. Though dry mud may break in a different manner than an artifact, the generated examples seem sufficiently similar and challenging. Archaeologists we consulted with preferred this procedure to other alternatives.

Figs. 9,10 demonstrate our re-assembly for several such examples. Both the number of fragments and the shape of the patterns vary. Color fading is due to the artifacts themselves. Our algorithm is shown to perfectly reassemble the vast majority of these artifacts, while missing only a few matches in the others.

7.2. Quantitative results

The most common measure used for square-pieces puzzles of natural images is the *average neighbor* [3,12–15]. It reports on the fraction of correct pairwise adjacencies. Since in our case the transformation space is continuous (rather than having only 16 possible transformations), we suggest a modification of the popular neighbor measure. It should take into account not only the adjacency of the pairs, but also the accuracy of the transformation. Thus, we define the *continuous-neighbor measure* as the fraction of the “correct” adjacent fragment pairs. Adjacency is correct if it satisfies the condition that the intersection fraction between the relative transformation of the pair and the relative ground truth transformation applied to each of the fragments, is larger than a given threshold ($\frac{2}{3}$ in our setting).

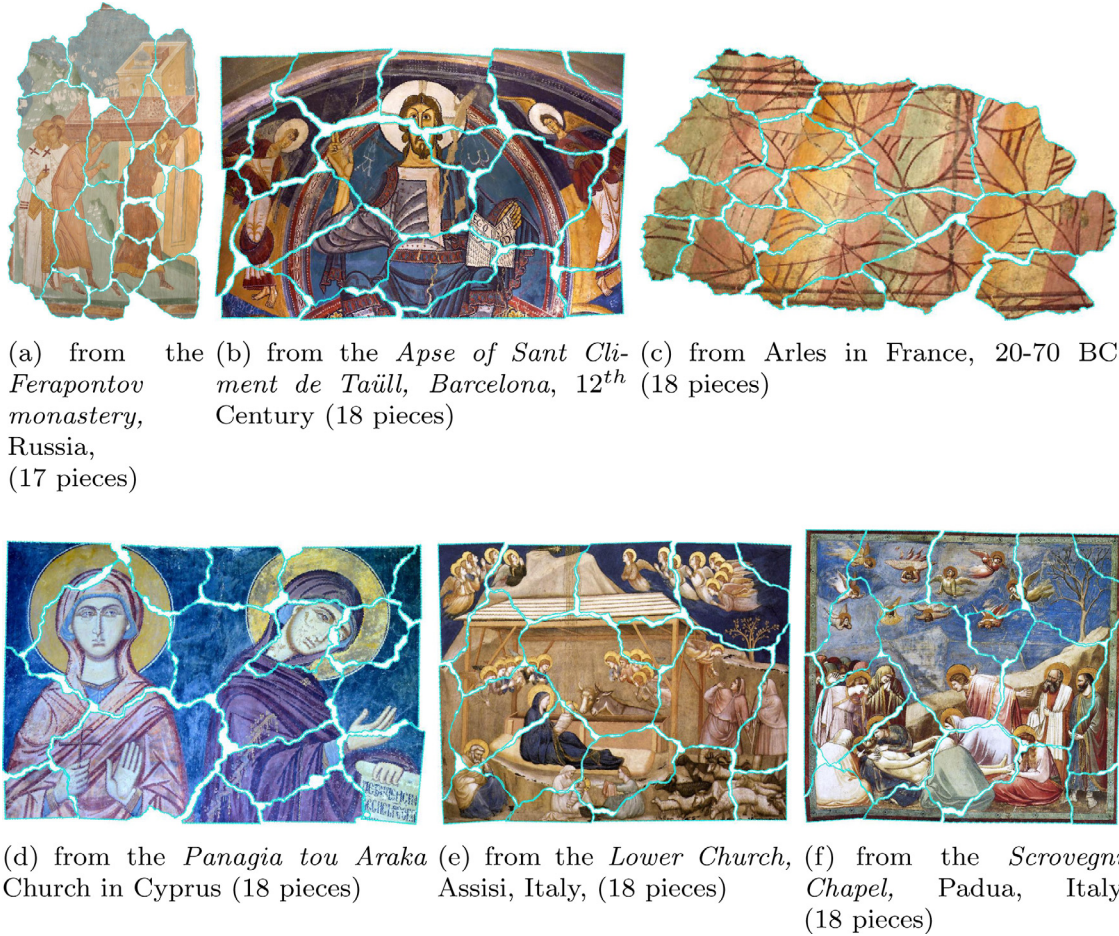


Fig. 9. Qualitative results. The geometric partition of the frescoes varies, as well as their patterns and the colors. Some frescoes have many repeating patterns, which make them more difficult to solve; some have only a few colors that occupy large regions; others have a larger variety of colors. Our algorithm managed to reassemble these examples flawlessly.

Table 1
Quantitative results.

# pieces	# artifacts	neighbor
≤ 10	8	100%
11 – 20	19	94.5%
21 – 30	6	99%
31 – 40	9	95.6%

Table 1 reports on the results of this continuous-neighbor measure. The results are averaged over all examples with the specified number of fragments. It is evident from the table that the results are almost perfect.

7.3. Implementation

The algorithm was implemented in MATLAB. It was run on i7-7700 with 16GB-RAM.

The running times vary between 6 minutes for 11-piece puzzles to 200 minutes for 36-piece puzzles. The most time-consuming component of the algorithm is the re-assembly, since it requires dissimilarity re-calculation of the pieces against the partially-reconstructed puzzle. For instance, in the example from the Panagia tou Araka Church in Cyprus (26 pieces), shown in Fig. 10(c), the reassembly takes 68 minutes, computing the initial confidence takes 8 minutes and the dissimilarity computation takes 32 minutes. The size of each fragment has almost no influence on the

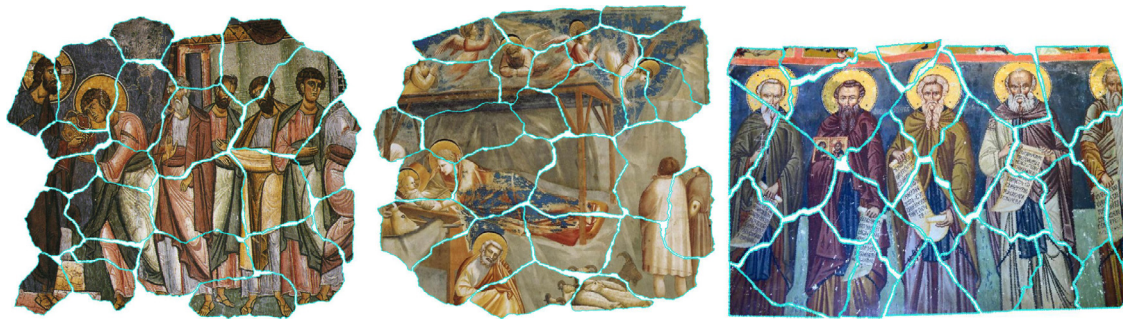
computation time, since the pieces are normalized, as discussed next.

As for the input: The size of each fragment image is 250×250 pixels on average. We note though that each fragment has a different size. Furthermore, great precision is not needed when taking pictures of the fragments, as compatibility is measured in terms of colors and not of geometry. However, it is required that the pictures of all the fragments be taken at the same conditions in terms of the distance of the camera, the direction of the camera, and the lighting conditions. Otherwise the patterns and colors might not match. There are algorithms that may be applied to images that are taken at non-similar conditions (such as white balancing), but it is recommended to avoid using them. Note that the orthographic projection is not required.

Finally, the fragments should be photographed on a background whose color differs from the colors on the fragment's boundaries, in order to be easily extracted. Extraction is performed by applying the palette algorithm, described in Section 5.1, determining the pixels whose colors are the most similar (L_2 -wise) to the colors of the pixels in the image corners.

7.4. Limitations

Fig. 11 (a) shows an example for which our algorithm fails to perfectly reassemble a fresco. Since this fresco contains a large gap in the middle, neighbors on both sides of the gap are not matched. Rather, matches that continue the patterns were selected. Never-



(a) from the *Asinou Church*, (b) from the *Scrovegni Chapel*, Padua, Italy, 1305, Church in Cyprus (26 pieces)
Nikitari, Cyprus, (23 pieces)



(d) from the *Agios Nikolaos* (e) from the *Scrovegni Chapel*, Arena Chapel, Agiasmati Church in
tis Stegis church in Cyprus (26 pieces) Padua, 1304, (32 Padua, 1305, (32 Cyprus (36 pieces)
pieces)

Fig. 10. Qualitative results, Cont'.



(a) Limitation 1: left; ground truth, right: reassembly result (b) limitation 2:

Fig. 11. Limitation. (a) The algorithm successfully reassembled the pieces on both sides of the large white gap, but failed to find the correct matches across the gap. (b) The piece on the left was incorrectly placed.

theless, the neighbor measure is still quite high (0.94). Fig. 11(b) shows another limitation, in which the piece on the left found a better match than the correct one, due to accidental color change on the boundary. Furthermore, as there is no strong constraint on geometrical information, pieces of flat colors without texture may be misplaced at the final stages of placement. Note that the same fresco, but with a different number of pieces, is shown in Fig. 10(g), where the reconstruction is perfect. This is due to the fact that the pieces have textures (the halo of the saint in this case), rather than a smooth color, and are thus easier to match.

8. Conclusion

This paper introduced a novel approach for re-assembly of an archaeological artifact from its fragments. It solves puzzles whose pieces are of general shape, whose boundaries are abraded, and whose colors are damaged. To deal with these challenges, our algorithm is based on four key ideas.

First, prior to re-assembly, each fragment is extrapolated, predicting how it should be continued by its adjacent piece. This re-

duces the problem of continuity of the fragments to the problem of matching them. Second, valid transformations between two fragments are defined as transformations for which one fragment does not intersect the other, but does intersect its extrapolation. Using the concept of *configuration space*, borrowed from robotics, allows us to compute the valid transformations efficiently. Third, a dissimilarity score is developed, which addresses the problems arising from degradation, as well as from the imprecision of the transformations due to sampling. Fourth, the confidence in the match is computed, by choosing the most relevant competitor and comparing their dissimilarities. This prevents early mismatches of the greedy placement.

Our algorithm was evaluated on a variety of archaeological broken artifacts and frescoes. It was shown, both qualitatively and quantitatively, to faithfully reassemble them. The reports on this work in the popular media can be found in [63]. A link to the Code will be added to this web site upon acceptance.

There are several interesting and challenging extensions to this work. First, this method may be extended to 3D objects, for reconstruction of painted pots, vases, plates etc. The main difficulty

is that a general 3D reconstruction requires estimating 3D transformations, which are of higher dimension and therefore, sampling the transformation space is far more challenging. Second, one may consider applying the method to text fragments. This requires more careful consideration for continuation, as there are no large portions of colors, but rather thin curves (text) that must be continued precisely. Finally, if one seeks for full reconstructions, our method can be combined with inpainting techniques. Note that our extrapolation may be used for this, however it is not as sharp as modern in-painting techniques such as for example [64].

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was funded by the GRAVITATE project under EU2020-REFLECTIVE-7-2014 Research and Innovation Action, grant no. 665155, by the Israel Science Foundation (ISF) 1083/18, and by the Ministry Science and Technology (MOST) 3-17513.

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